



Crack observation methods, their application and simulation of curved fatigue crack growth

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ABSTRACT

The observation of cracks with curved crack path has to be done optically. It is shown that a modified commercial flat bed scanner and a combination of a high-resolution scanner camera system with $10,000 \times 10,000$ pixels with a telecentric lens are appropriate for high-resolution (up to $8 \mu\text{m}/\text{pixel}$) optical recording of multiple crack ends on sample areas up to $210 \text{ mm} \times 290 \text{ mm}$. The high-resolution photographs are suitable for determination of crack lengths. It is also possible to observe crack paths or geometrical crack tip parameters and strain fields by image correlation. The method is used to determine static crack resistance and cyclic crack growth curves on center crack tension and biaxial cruciform samples.

Furthermore, the paper presents an improved finite element technique for the simulation of curved fatigue crack growth in a multiple arbitrarily pre-cracked isotropic sheet under biaxial plane stress loading applying a predictor–corrector procedure in combination with the modified virtual crack closure integral (MVCCI) method including the consideration of the plastic limit loads. For this, the program PCCS-2D was extended to analyse the crack growth and the plastic limit load for each crack propagation step in a fully automatic simulation. The proposed solution algorithm provides a powerful tool for flaw assessment with the failure assessment diagram procedure in combination with a numerical crack path simulation. Finally, the simulation is verified experimentally.

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1. Introduction

For fracture mechanics investigations it is important to measure the propagation rate and direction of cracks. The direct optical observation of the crack is the safest measurement method to determine the position of the crack tip and to calculate the crack length. Nevertheless, it can hardly be automated and the detection of small changes in crack size needs high magnification. Indirect crack observation methods were developed in the past. Among these the potential drop method where the electric potential over the crack tip is measured while a constant current is sent through the sample is widely used. Another common method is the measurement of small changes in stiffness of the sample (compliance method). These methods

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Nomenclature

a	crack length
a^l	crack length of the l th crack
A_5	strain to failure
A_{50}	strain to failure
A_g	uniform elongation strain
C	material constant of the Paris–Erdogan equation
F_{max}	maximum load during cyclic loading
F_{min}	minimum load during cyclic loading
F_X	load in X-direction
F_Y	load in Y-direction
$J_{i, BL}$	fracture J -integral at initiation of stable crack extension (intersection with the blunting line)
K_I^l	mode I stress intensity factor (l th crack)
$K_I^{l, max}$	mode I stress intensity factor of the maximum loaded crack
K	stress intensity factor
K_I, K_{II}	mode I and mode II stress intensity factors
K_{IC}	plane strain fracture toughness
K_{eq}	equivalent stress intensity factor
$K_{J, i, BL}$	converted fracture J at initiation of stable crack extension
K_{mat}	fracture toughness (resistance for FAD)
K_{max}	maximum stress intensity factor during cyclic loading
K_r	FAD parameter
l	crack number index
l_{max}	number of the crack tip with the maximum stress intensity factor
L_r	FAD parameter
M	total number of cracks
n	material constant of the Paris–Erdogan equation
N	loading cycles
R	loading ratio
R_m	ultimate tensile strength
$R_{p0.2}$	yield strength
δ_5	crack opening displacement over a 5 mm gauge length at tip of fatigue pre-crack
$\delta_{5,0.2, BL}$	crack opening displacement at initiation of stable crack extension (intersection with the 0.2 mm shifted blunting line)
$\delta_{5,i, BL}$	crack opening displacement at initiation of stable crack extension (intersection with the blunting line)
Δa	stable crack extension
Δa^l	crack extension increment (l th crack)
Δh^l	virtual crack extension increment (l th crack)
$\Delta h^{l, max}$	maximum virtual crack extension increment
ΔK	cyclic stress intensity factor range
$\Delta \bar{K}_{II}^l$	virtual mode II stress intensity factor increment (l th crack)
ΔN	increment of loading cycles
$\Delta \varphi^l$	slope at the end of the curved crack extension increment (l th crack)
$\Delta \lambda^l$	ordinate at the end of the curved crack extension increment (l th crack)
λ	biaxial load ratio
σ	stress
σ_e	yield stress (resistance for FAD)
σ_{min}	minimum stress during cyclic loading
σ_{max}	maximum stress during cyclic loading
Ψ	crack tip opening angle
Ψ_C	critical crack tip opening angle
2D	two-dimensional
APDL	Ansys Parameter Design Language
CCD	charge-coupled device
CTOA	crack tip opening angle
CTOD	crack tip opening displacement
EDM	electro-discharge machining
FAD	failure assessment diagram
MVCCI	modified virtual crack closure integral
PCCS	predictor–corrector crack simulation

are well established for standard tests in particular because their signals are recordable easily. Each method has its own additional sources of error – for instance the local change of electric conductivity by plastic deformation. In addition these indirect crack length measurement methods give only a projection of the crack length. Changes in crack direction are not determined. Therefore, most standards require a calibration of these methods with optical measurements and give restrictions for the tolerable change of crack direction, see e.g. [1].

The direct optical crack observation is usually done by a (video) microscope which is moved by linear actuators with integrated length measurement systems. For the measurement the test is paused, the crack tip positions are located within the microscope picture. With this method it is possible to observe cracks on curved paths [2]. The main handicap of the existing practice is the small image section which can be viewed in high magnification. For the observation of crack propagation which is longer than the object field of the microscope, the position of the microscope relative to the specimen must be changed. In [3] a line pattern on the specimen surface is used to identify the absolute position of the picture relative to the sample. In complex cases of crack path observations it is important to detect the initiation of additional cracks while other cracks propagate. If it is not possible to observe the full measurement area automatically, an operator must be present during the whole test and data is lost if additional cracks are not recognized. To overcome this, it is proposed to take full-field high resolution pictures of the complete measurement area of the sample. CCD-line scanners are competitive tools for this purpose if still image conditions can be achieved. Within such a picture it is possible to measure the crack position as well as geometrical crack tip parameters like the crack tip opening angle Ψ (CTOA) or δ_5 which are used especially for testing plates with low constraint [3–5]. Instead of tactile δ_5 clip on gages for instance the displacement information of the image can be used. There are several sophisticated optical methods like caustic, photoelasticity or moiré interferometry to obtain strain fields near the crack tip with high resolution [6]. All of them need additional equipment. Photogrammetry in the version of digital image correlation allows the usage of the same pictures which are needed to obtain the crack paths. It was widely applied to fracture mechanics questions [7–9]. Pictures at deformed states are compared with a reference. For that purpose the picture is divided in subsets and the transformation (displacement, rotation, deformation) of the subset is determined. The accuracy is higher than the pixel size because the greyscale information of several pixels is used together. There is only the need to apply a greyscale pattern on the sample surface which gives enough contrast information in the subsets. In the present study this is done by application of a stochastic graphite spray pattern. The high resolution of the scanning devices is necessary for the detection of details on the samples surfaces. However, the high number of pixels does not give proportional higher precision in deformation measurement because of the movement of the CCD line. In [10] the accuracy of a camera scanner was determined to 1/7 pixel while a camera with fixed CCD area chip reached 1/30 pixel.

The application of the full field high resolution CCD-line scanner devices gives a benefit for all plane specimens where long crack paths must be observed. The camera scanner system was especially designed to observe the complete measurement area of planar biaxial samples on which additional cracks often form during testing. The numerical simulation of crack propagation is used to accompany the cyclic tests on planar biaxial specimens. It gives the possibility to study the crack propagation like it is found in structures under several planar stress states on a clearly arranged sample.

In addition to the experimental determination of crack propagation, the results of crack propagation simulation have direct relevance for the safety as well as the non-destructive testing of real structures. Several numerical methods like the finite element method (FEM), the extended finite element method (XFEM), the boundary element method (BEM) or various meshless methods were successfully applied for the simulation of crack propagation [11–20] by using a step-by-step approach. The predictor–corrector method (PCM) [2,19] was proven to be very accurate and robust in computing single cracks and multiple crack systems, even in the case of very coarse meshes. In contrast to the mostly applied straight crack extension increments the PCM uses a higher order simulation process with a piecewise curved approximation of the crack extension increments. As an important advantage of the method, less crack propagation steps are required without a loss of accuracy which reduces the computational cost significantly. The results of the simulations are usually the histories of the stress intensity factors as a function of the changing crack geometry and the given loading system. To carry out flaw assessment with the failure assessment diagram procedure in combination with a numerical crack path simulation a complementary numerical analysis of the plastic limit load at each step of the simulation is necessary. For this reason a robust method is developed in order to combine advanced crack path simulation with the flaw assessment using the failure assessment diagram. A program based on the commercial FEM code ANSYS is developed to solve general problems full automatically without any user interaction.

The aim of this paper is to present improved experimental techniques to obtain full-field high resolution images in large planar areas together with the numerical evaluation of the measured curved crack paths by the predictor–corrector method as well as the plastic limit load by the lower bound theorem of plasticity.

2. Materials and methods

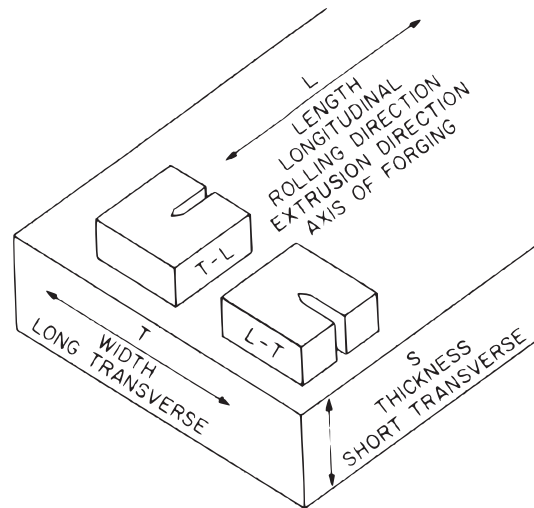
2.1. Materials

The experiments were carried out on the aluminium alloys EN AW-Al Zn6CuMgZr (7050) in the state T7451 and EN AW-Al Mg1SiCu (6061) in the state T651, respectively. The mechanical properties, i.e. yield strength $R_{p0.2}$, ultimate tensile strength R_m and uniform elongation strain A_g as well as the strain to failure A_{50} and A_5 , respectively, are given in Table 1.

Table 1

Tensile properties of the aluminium alloys 7050 T 7451 and 6061 T 651, tested according to DIN EN 10 002.

Material	Direction	$R_{p0.2}$ (MPa)	R_m (MPa)	A_g (%)	A_{50} (%)	A_5 (%)
7050	Longitudinal	464	520	7.3	12.7	–
7050	Transversal	463	534	6.8	10.3	–
6061	Longitudinal	286	313	1.2	–	1.9
6061	Transversal	274	298	1.2	–	2.1

**Fig. 1.** Test directions according to ASTM E399.

The mechanical properties parallel and perpendicular to the rolling direction show only small deviations. The rolling and testing directions of the sheets are represented as L–T and T–L as defined in ASTM E399 (see Fig. 1).

Standard fracture mechanics properties of the two investigated alloys were determined for reference. The plane strain fracture toughness of the 7050 alloy was measured on CT specimens with 25.4 mm thickness according to ASTM E 399 to $K_{IC} = 36 \text{ MPa}\sqrt{\text{m}}$ in the L–T and $K_{IC} = 30 \text{ MPa}\sqrt{\text{m}}$ in the T–L direction. The ductile crack initiation of the 6061 alloy was measured according to ISO/FDIS 12135 on single-edge notched bending samples of 19 mm thickness to $J_{i\text{BL}} = 41.1 \text{ kJ/m}^2$ for the L–T direction and $J_{i\text{BL}} = 31.4 \text{ kJ/m}^2$ in the T–L direction, respectively.

2.2. Modified flat bed image scanner

A commercial flat bed scanner with CCD line sensor *CANON Canoscan 4200F* which gives a resolution up to 3200 dots per inch was modified for the high magnification observation of areas up to $210 \text{ mm} \times 290 \text{ mm}$. The modifications consist mainly of the removal of the transmitted light unit together with the cover. For the vertical usage reinforcements of the body with fixing holes were adapted. The frame of the glass panel was changed that the latter can contact directly the specimen. The distortion was determined by measuring a sheet of millimetre paper against the coordinate measurement table to be smaller than 0.02 mm. The application of the flat bed scanner on a center crack tension sample is shown in Figs. 2 and 3b.

2.3. Camera scanner system

Especially the insufficient depth of focus limits the usage of the flat bed scanner system for instance, if anti-buckling devices are used or if samples for biaxial crack growth tests with reduced thickness in the middle area are investigated. To enhance the field of application for such cases a system with a camera scanner *Pentacore SCAN 6000* is applied. It consists of a 40 mm trilinear CCD line which is moved over an image field of $40 \text{ mm} \times 40 \text{ mm}$. Pictures with a resolution of up to $10,000 \times 10,000$ pixels can be taken from this area. The scan time for the whole image field is approximately 120 s. An extensive telecentric lens is used to get an object field of $120 \text{ mm} \times 120 \text{ mm}$ with negligible distortion for a lens-specimen distance of 400 mm. The depth of focus is approximately 4 mm. The resolution of this system was measured with a reticule of lines to 45 line pairs per millimetre. This is in the range of the projected pixel size on the object side. Fig. 3a shows the camera system together with the laser based alignment and positioning kit and the LED ring light. With this system

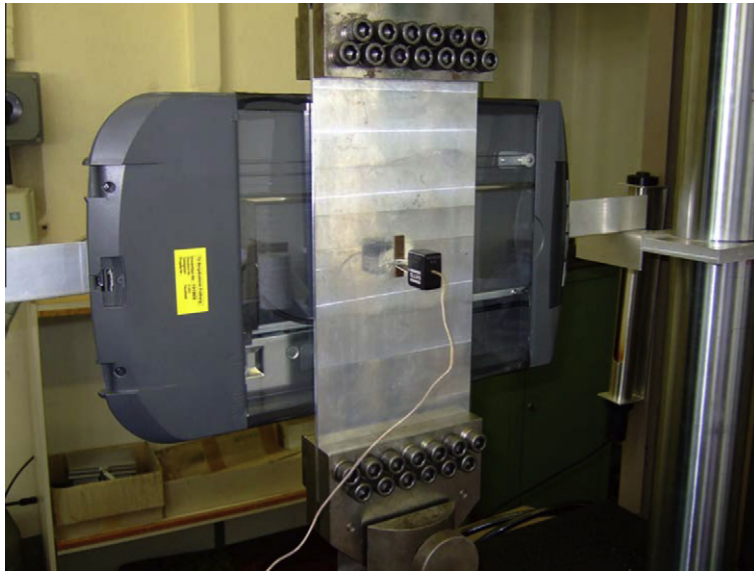


Fig. 2. Test set-up with modified flat bed scanner and 160 mm center crack tension sample.

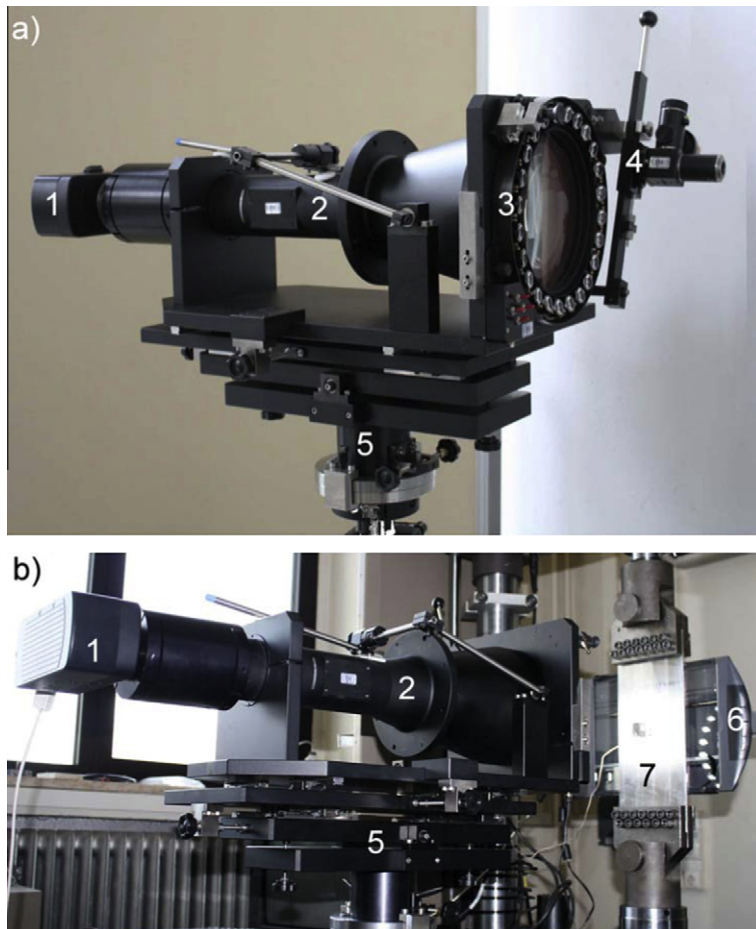


Fig. 3. (a) Scanner camera system, (b) Flat bed Scanner and camera system applied to a 160 mm center crack tension sample for measurement of crack resistance curves, (1) Pentacon scan 6000, (2) telecentric lens, (3) LED ring light, (4) laser collimator for alignment, (5) alignment fixture, (6) flat bed scanner, (7) center crack tension sample.

the incident light goes nearly orthogonal to the specimen surface. The telecentric lens selects mainly the light reflected in the backward direction again. Thus, all cracks appear black.

2.4. Measurement using center crack tension specimen

The tests were performed using standard center crack tension samples of the dimensions $400 \text{ mm} \times 160 \text{ mm} \times 5 \text{ mm}$ made of the aluminium alloy 7050 with a 3 mm electro-discharge machined (EDM) starting notch on a Servo-Hydraulic Universal Testing Unit MTS 880. The plates were milled out of a 80 mm thick sheet.

Fig. 2 shows the test set-up. Crack growth curves were measured in sine shape load-control with a constant stress range of 90 MPa and a stress ratio $R = 0.1$ at a frequency of 3 Hz. The measurement areas of the samples were polished mechanically. Additional pre-cracking was not performed. When the calculated crack extension from the compliance measurement extended 0.7 mm or the cycle counter exceeded 2000 cycles, relatively to the last picture, the test was interrupted and a picture of 3200 dpi resolution of the defined measurement area ($165 \text{ mm} \times 32 \text{ mm}$) was taken, before the test was resumed. The freeware script interpreter *runcmd* was used to automate the control of the scanning software. With this script the sample identification and the actual cycle counter or servo cylinder stroke were included in the file name of the picture, too. This procedure runs automatically without the need of an operator. The analysis of the pictures is done offline after the test.

For the measurement of crack length the free available JAVA based image processing system *ImageJ* was used. For that purpose an *ImageJ*-plug-in was written which opens the pictures, starting with the longest crack length for better recognizability of the crack tip. In this image the user must select the crack tips to measure. The analysis is then done backwards to shorter crack lengths. The plug-in shows automatically the zone of the crack tip on a high magnification which shows all pixels of the image. The same area of two subsequent images is presented to the operator simultaneously to compare the crack propagation. Only the crack tips have to be marked by hand. The output is a xy -coordinate table of the crack tips and crack length related to the cycle counter. Usually, sequences of 50–100 images are produced during one test. It takes the user about 30 min to mark all crack tips for one sample in all pictures. In principle the automated trace of the crack tips is feasible by an image recognition software. Up to now it was not realized. Due to the high image size sophisticated data reduction technology would be needed to achieve proper computing time and high accurateness.

The measurement of static crack resistance curves was done in a similar way. A notch of length $2a = 40 \text{ mm}$ was machined by electro-discharge machining (EDM). To obtain a sharp crack tip with a small plastic zone the sample was pre-cracked at $R = 0.1$ by stepwise fatigue load reduction to a value of $\Delta K = 8.6 \text{ MPa}\sqrt{\text{m}}$ in accordance to ISO 22889 [4]. Then a stochastic point pattern was applied on both sides of the sample surface using graphite spray to get sufficient contrast for an image correlation measurement. The average size of the dots of the pattern was $60 \mu\text{m}$ with a 10%-quantile of $25 \mu\text{m}$ and a 90%-quantile of $85 \mu\text{m}$. During the static test a tensile load was applied by cylinder stroke control. On one side of the sample the flat bed scanner was mounted, from the other side the scanning camera system was used (Fig. 3b). Every 0.1 mm stroke the test was stopped to take two images of the measurement area. The crack extension was measured with *ImageJ*. The 2D version of the image correlation software *ARAMIS* from the company *GOM* was used to measure the deformation field within the high resolution pictures of both measurement devices. Geometrical crack tip parameters like δ_5 were determined from these strain field measurements.

2.5. Crack path in planar biaxial specimen

Cyclic biaxial crack growth experiments were performed on a cruciform slotted sample similar to [21] with a thinner inner measurement region with 2 mm thickness and approximately $120 \text{ mm} \times 120 \text{ mm}$ useable measurement area (Fig. 4). The sample was machined from a rolled sheet of the 6061 alloy with 19 mm thickness by CNC milling. The starting notches were inserted by EDM as two holes with diameter 10 mm on the positions $(-15 \text{ mm}, 15 \text{ mm})$ and $(15 \text{ mm}, -15 \text{ mm})$ with an additional slit of 10 mm length at each hole in X -direction (see Fig. 4). The resulting notch radius was 0.25 mm. There was no additional pre-cracking of the shown sample. For a better detectability of additional cracks no point pattern for digital image correlation was applied. The investigations were done on a servo-hydraulic 250 kN *INSTRON* biaxial tension compression machine. The ratio between the loading axes was $\lambda = F_Y/F_X = 0.5$. The load ratio was chosen as $R = F_{\min}/F_{\max} = 0.3$ because in previous investigations [2] this value was found to give no crack path deviations by crack surface roughness induced friction on this material. The test was done using synchronous sine shape load with a maximum of 81 kN in X -direction at a frequency of 10 Hz. This load is equal to an initial maximum stress intensity factor $K_{\max} = 18.7 \text{ MPa}\sqrt{\text{m}}$.

2.6. Crack path simulation

In the program code *PCCS-2D* (predictor–corrector crack simulation) the numerical simulation of the geometrical evolution of cracks is based on a well-founded fracture mechanics predictor–corrector method on the basis of the criterion of the local symmetry. Furthermore the implemented simulation concept fulfils the geometrical conditions on crack growth by the use of a quadratic form function for the increments. For a system of a number of M growing cracks using compatible Δh^l ($l = 1 \dots M$) the following geometrical quantities for the incremental steps of the simulated crack trajectories as shown in Fig. 5 are obtained [19]:

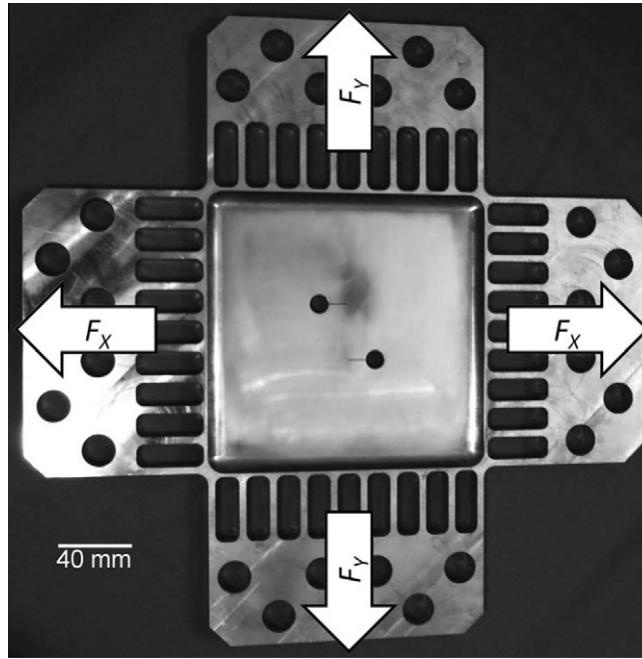


Fig. 4. Planar biaxial sample with starting notches.

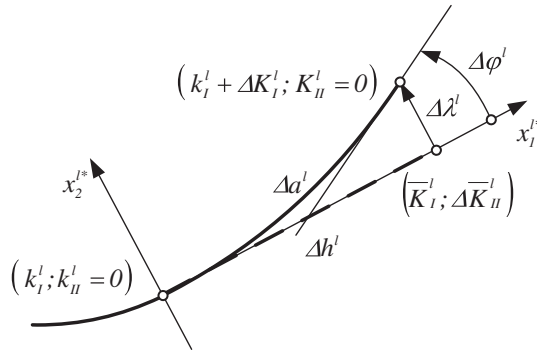


Fig. 5. Predictor-corrector step of crack path simulation.

$$\Delta \varphi^l = -2 \frac{\Delta \bar{K}_{II}^l}{k_I^l}, \quad \Delta \lambda^l = -\frac{\Delta \bar{K}_{II}^l}{k_I^l} \Delta h^l, \quad \Delta a^l \approx \Delta h^l \left[1 + \frac{2}{3} \left(\frac{\Delta \bar{K}_{II}^l}{k_I^l} \right)^2 - \frac{2}{5} \left(\frac{\Delta \bar{K}_{II}^l}{k_I^l} \right)^4 \right]. \quad (1)$$

The maximum amount of virtual crack extension increment $\Delta h^{l \max}$ should be taken for the crack tip $l_{\max}$, where the maximum stress intensity factor $k_I^{l \max}$ occurs. Using the Paris law with a chosen number of loading cycles ΔN we obtain:

$$\Delta h^{l \max} = C \left[k_I^{l \max} (1 - R) \right]^n \Delta N, \quad R = \frac{\sigma_{\min}}{\sigma_{\max}}, \quad \Delta h^l = \Delta h^{l \max} \left(\frac{k_I^l}{k_I^{l \max}} \right)^n. \quad (2)$$

By considering Eq. (1) it can be clearly seen, that the simulation of crack growth requires highly accurate stress intensity factors in order to reduce numerical errors in the computation of the crack extension increments. For this reason an improved MVCCI method [19] was developed, which provides a highly accurate computation of the separated strain energy release rates and the related stress intensity factors without special techniques also in cases of coarse meshing.

For the numerical simulation of the crack paths the program PCCS-2D based on the FEM code ANSYS in Ansys Parameter Design Language (APDL) is developed, which provides an automatic strategy and does not require any user interaction during the calculation. According to the user defined input data the program involves the cracks in the geometry by using the Boolean operations. A mapped mesh of chosen size is defined around the cracks while the rest of the geometry is meshed

freely. This ensures a mesh with high quality around the cracks and an automatic meshing strategy for general geometrical configurations. In the present FEM analysis, quadratic elements of the type PLANE82 are chosen. In the program the crack trajectory is computed based on curved segments in a predictor–corrector method instead of the widely used straight segments. This decreases the computational effort as the number of crack propagation steps can be reduced but requires a local re-meshing. It should be mentioned that additional cracks can be implemented at any step of the simulation by an update of the geometry.

In a failure assessment diagram (FAD) an assessment point with the parameters $K_r = K/K_{mat}$ and $L_r = \sigma/\sigma_e$ is compared with a failure line, which is calculated from the tensile material properties. K is the stress intensity factor, K_I in a mode I case. In cases with initial mixed mode loading the equivalent stress intensity factor K_{eq} should be taken. The tensile stress σ stands for the loading of the component and σ_e is the corresponding plastic limit load.

As mentioned above, for the practical use of the failure assessment diagram, it is required to calculate the L_r parameter as a degree of plasticity effects. The global plastic limit load σ_e has to be calculated for an elastic–perfectly plastic material

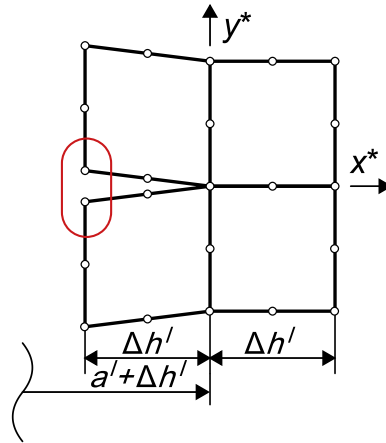


Fig. 6. Used node pair of the established finite element mesh.

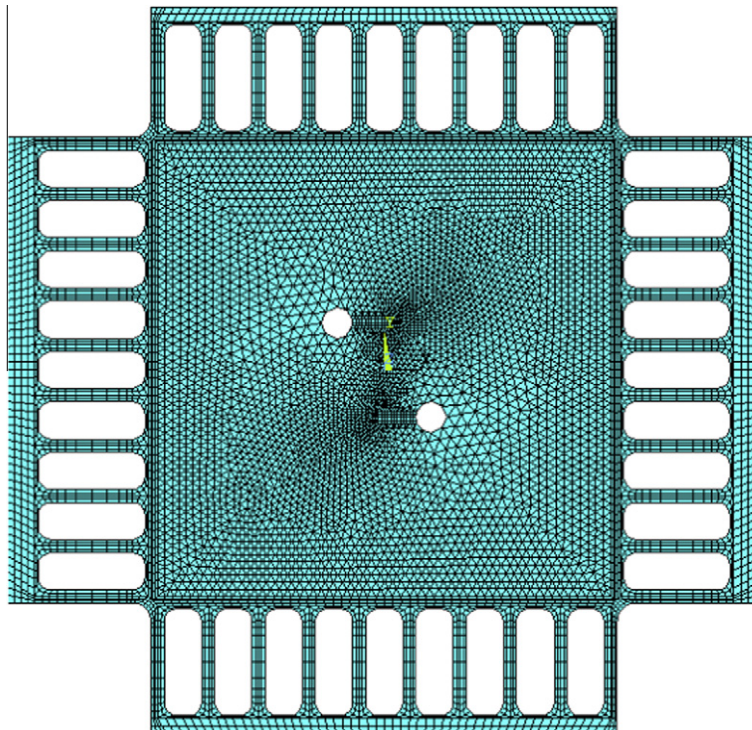


Fig. 7. Finite element model of the planar biaxial specimen with crack length $a = 10$ mm (ANSYS).

model with the lower bound theorem of plasticity [22]. Only primary loads are taken into account. Based on the given modular concept of PCCS-2D a simple iterative solution strategy for proportional loading systems is developed within the FE-code ANSYS. For the given 2D problems plane stress and von Mises plasticity are considered. The plastic limit load is evaluated from the characteristic non-linear load–deformation curve. During the growth of plastic deformations, the slope of the non-linear load–deformation curve declines and finally closes to zero. In accordance to this the solution strategy is based on the slope evaluation of the elastic and plastic region as well as the use of a plastic slope criterion. For the numerical calculation of the plastic limit load it is assumed, that the crack tips are the most loaded locations in the cracked specimen. Thus, it is necessary to analyse the deformation and evaluate the slope of the load–curve at this locations. Based on the established finite element mesh for the stress intensity factor calculation with the MVCCI method, a node pair near the crack tip l_{max} with the maximum stress intensity factor $k_I^{l_{max}}$ is chosen as the deformation parameter. The distance of the node pair to the crack tip is constant Δh^I (Fig. 6). In an unloaded specimen condition, these nodes have the same coordinates. By increasing the load the displacement of the nodes is raising. With the evaluation of the displacements and the displacement differences for the node pair it is possible to calculate the load–displacement slope for the elastic and plastic case.

The numerical calculation of the plastic limit load is executed in the course of the stress intensity factor calculation (linear elastic simulation) for a given crack condition, where the elastic slope for the loaded specimen will be evaluated. After this an analysis of the same crack condition with elastic–perfectly plastic material behaviour follows. In this analysis the load of the cracked specimen changes in loops until the plastic slope criterion is satisfied. The plastic limit load for a given crack condition is detected, when its evaluated slope of the elastic region is between 20 and 24 times bigger than the plastic slope. The used simple iteration solution strategy for the numerical calculation of the plastic limit load was verified using some examples of [23]. The applied finite element model of the planar biaxial specimen is shown in Fig. 7.

3. Results

3.1. Crack growth curves

Fig. 8 shows an image of a 160 mm center crack tension sample with two cracks, each approximately 50 mm in length, and a magnification of the right crack tip from the same picture to demonstrate the high resolution. In a direct comparison there was no significant difference between the flat bed scanner measurements and a video microscope. Fig. 9 shows the

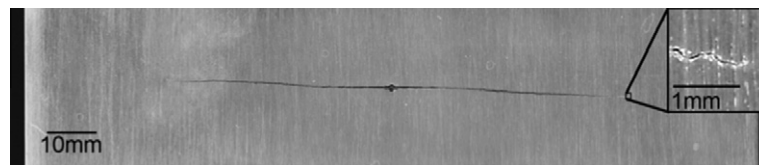


Fig. 8. Picture of the measurement area of a 160 mm center crack tension sample and area with higher magnification.

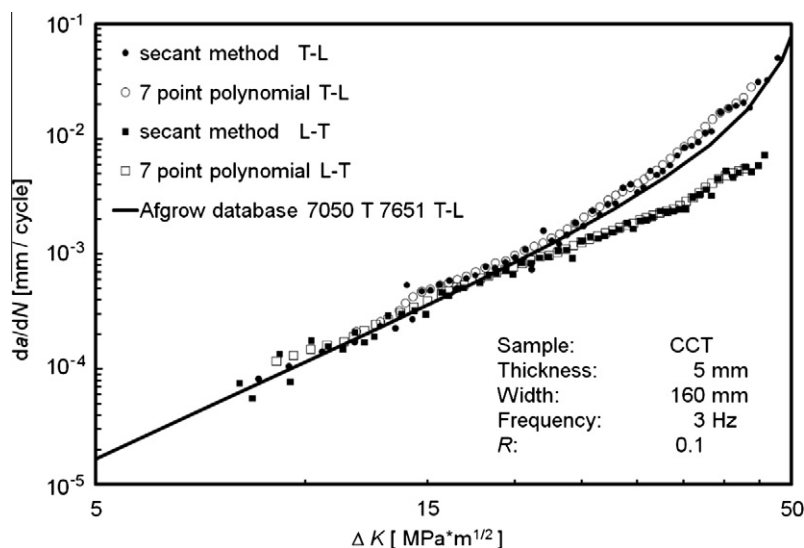


Fig. 9. Crack growth curves measured with a flat bed scanner system.

crack growth curves measured with this set-up. As in other investigations [24] the crack growth rate for T–L direction is significantly higher than in L–T direction for $\Delta K > 20 \text{ MPa}\sqrt{\text{m}}$. For comparison an additional curve based on material data from the AFGROW Database [25] for sheets and plates of aluminium 7050 T 7651 in T–L direction is given and shows good accordance. The typical scatter of fatigue crack propagation measurements with the secant method (filled symbols) was smoothed using the 7-point polynomial method according to ASTM E 647-08 [1] (open symbols).

3.2. Crack resistance curves

One of the obtained deformation fields from the image correlation is shown in Fig. 10 together with the stochastic pattern and the complete measurement area. In Fig. 10a the clip-on for the crack opening displacement measurement is visible in the center of the picture. The plastic zone can be easily recognized in the digital image correlated deformation pattern in Fig. 10c. The obtained $\delta_5 - \Delta a$ -crack resistance curve for the L–T-direction is shown in Fig. 11. According to [4] a power law was fitted to the measured points

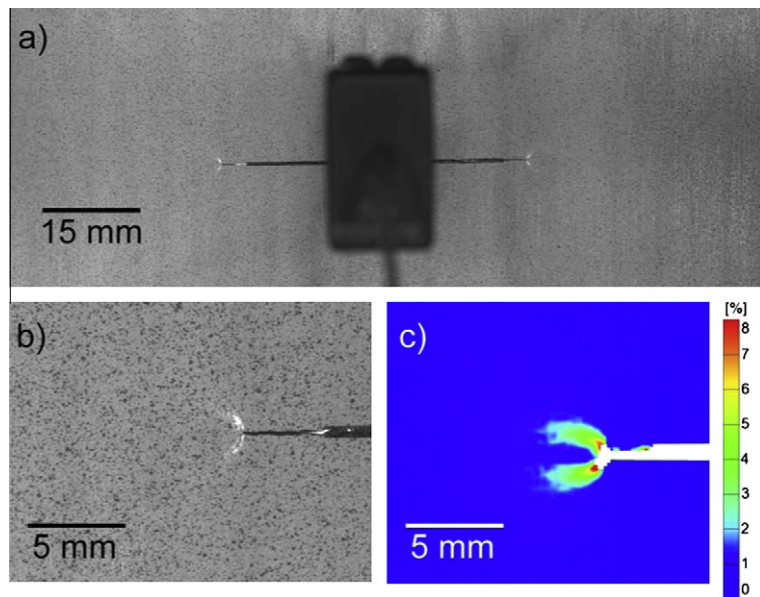


Fig. 10. Center crack tension specimen of aluminium alloy 7050 T 7451 L–T. (a) Measurement area, (b) stochastic pattern and (c) von Mises equivalent strain deformation field.

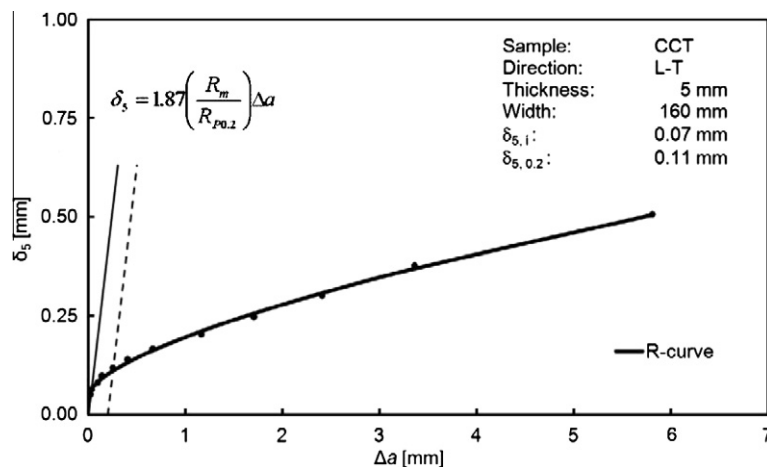


Fig. 11. Crack resistance curve of aluminium alloy 7050 T 7451 L–T.

$$\delta_5 = 0.05 + 0.145\Delta a^{0.647}. \quad (3)$$

A formal usage of the blunting line with

$$\delta_5 = 1.87 \frac{R_m}{R_{p0.2}} \Delta a \quad (4)$$

gives the initial values $\delta_{5,i \text{ BL}} = 0.07 \text{ mm}$ and $\delta_{5,0.2 \text{ BL}} = 0.11 \text{ mm}$. Both values are below the criterion of validity $\delta_{5,max} = B/30 = 0.17 \text{ mm}$ and therefore independent from sample geometry. Fig. 12 gives the values for the crack tip opening angle (Ψ) calculated from the δ_5 measurements. While there is some scatter it can be seen that the CTOA decreases from values higher than 30° to a constant level after about 1 mm of crack extension. The critical angle was determined as the mean value of all measurements between 1.2 mm and 6 mm stable crack extension to $\Psi_{\delta 5c} = 5.3^\circ \pm 1.6^\circ$. This value is in the range usually found for aluminium alloys: For an aluminium alloy 2024 T 351 values of the critical CTOA were measured by Mahmoud and Lease between 5.28° and 4.48° depending on the specimen thickness [26].

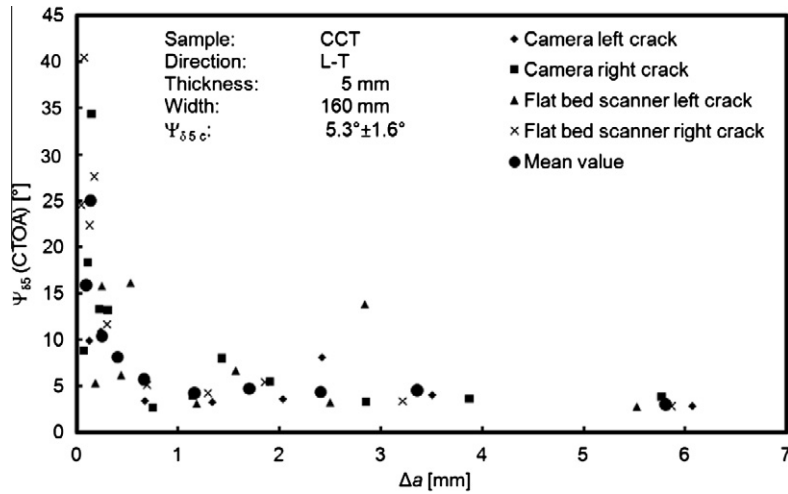


Fig. 12. Calculated CTOA values for aluminium alloy 7050 T 7451 L-T.

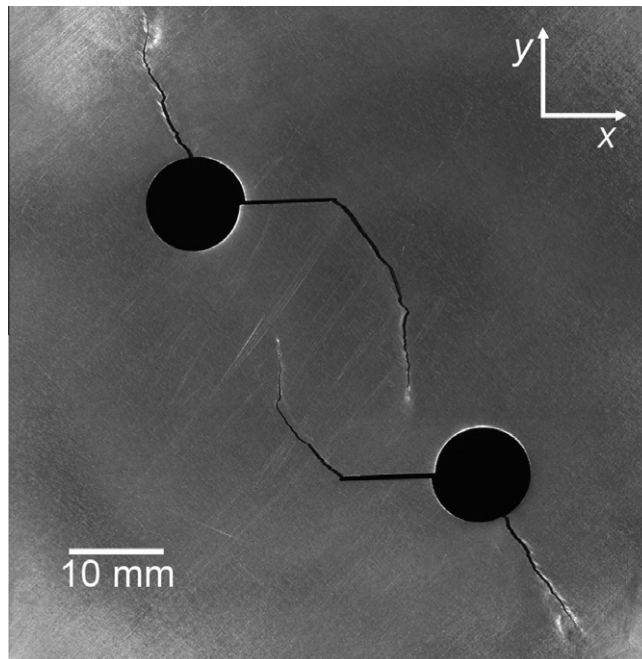


Fig. 13. Sample surface after 23,085 cycles for aluminium alloy 6061 T 651. $F_{X,max} = 81 \text{ kN}$, $F_{Y,max} = 40.5 \text{ kN}$, $R = 0.3$.

3.3. Complex crack growth in biaxial planar samples

The start configuration of the biaxial planar sample with two holes and two notches is given in Fig. 4. The sample failed after 23,098 cycles. Fig. 13 shows the surface of the sample after 23,085 cycles on the verge of fracture. Scanning the whole area of interest gave the advantage that the formation of two additional cracks after about 20,000 cycles could be observed at the holes when the two main cracks had a total crack arc length of 19.5 mm.

A comparison between calculated and measured crack paths in Fig. 14 shows that both are in principle accordance. The circles show the simulated crack path starting with the cracks 1 and 2. The newly initiated cracks were included in the crack path simulation as cracks 3 and 4 using a geometrical update of the model step when the cracks on the starting notches had a length of 19.5 mm. Starting at this step, the simulation points are marked as triangles in Fig. 14.

The real paths of cracks 1 and 2 start with a higher kink angle than the simulation. An additional test with the same conditions and 90° rotation of the rolling direction to the coordinate system of the sample showed in principal the same crack behaviour and achieved 24,480 cycles until final fracture.

The high amount of plastic deformation at the end of the test is demonstrated in the stress fields of the simulation for 22.5 mm crack length of cracks 1 and 2 in Fig. 15. The failure of the planar biaxial specimen was finally controlled by plastic collapse.

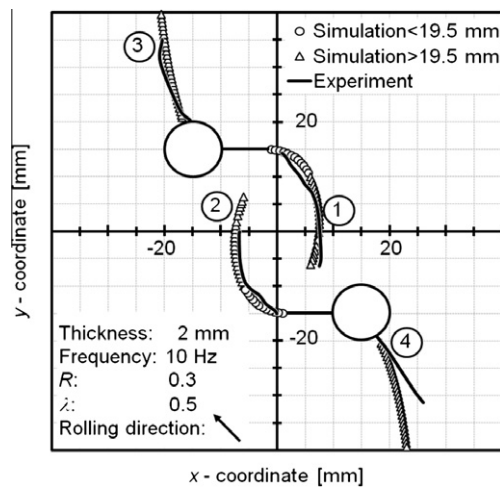


Fig. 14. Experimental and simulated crack paths in a cruciform sample of aluminium alloy 6061 T 651. The starting cracks 1 and 2 had a length of 10 mm. After crack growth of these cracks for additional 9.5 mm (i.e. a total crack length of 19.5 mm), the cracks 3 and 4 were initiated.

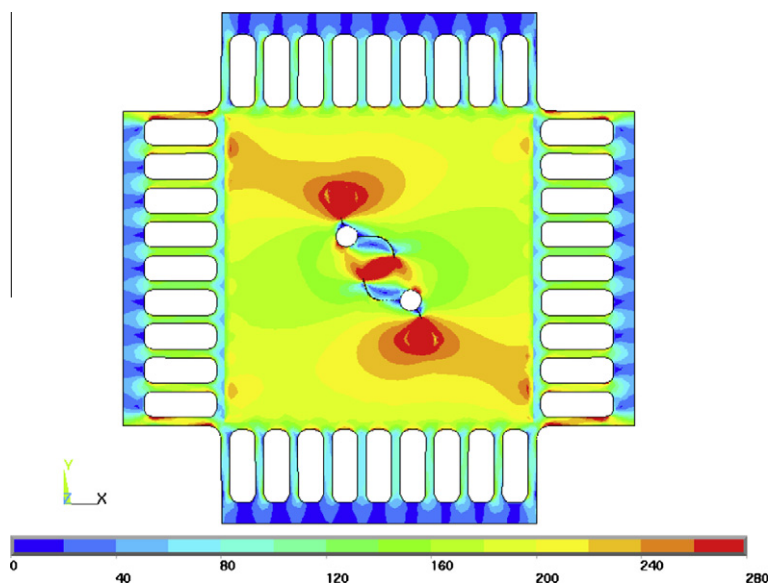


Fig. 15. Calculated von Mises equivalent stress at a crack length of 22.5 mm of cracks 1 and 2 (ANSYS).

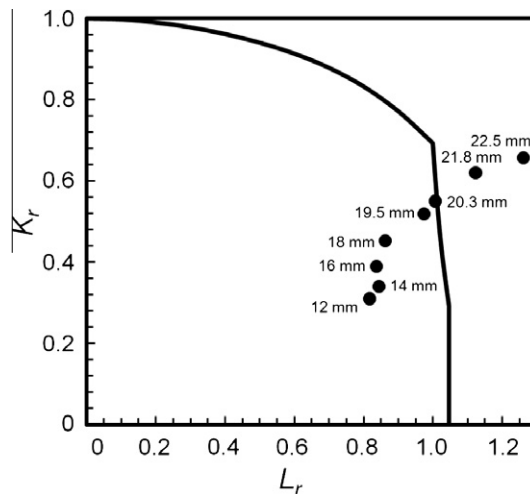


Fig. 16. Calculated failure assessment diagram with damage status of the sample at crack lengths of the starting cracks one and two included.

The K_{mat} for the failure assessment diagram was calculated from the lower value of $J_{i\ BL}$ in T–L direction to

$$K_{mat} = K_{J_{i\ BL}} = \sqrt{\frac{31.4 \text{ kJ/m}^2 \cdot 69 \text{ GPa}}{(1 - 0.3)^2}} = 66.5 \text{ MPa}\sqrt{\text{m}}. \quad (5)$$

The limit line was estimated according to the basic level of [23]. The calculated failure assessment diagram (Fig. 16) indicates the excess of the plastic limit load at a length of crack number 1 of 20.3 mm which is reached after approximately 20,500 cycles. The real failure occurred at a crack length of approximately 30.5 mm at 23,098 cycles by plastic collapse. This means that the calculated FAD is slightly conservative.

4. Summary

In this study, two techniques are presented to obtain high resolution images to determine the crack paths and deformation fields development on the complete measurement area of biaxial planar samples and 160 mm center crack tension specimen. Curved crack propagation can easily be observed automatically without the need of an operator during the tests due to the possibility to evaluate full-field high resolution images after the tests. There is no information loss if additional cracks occur and the complete primary data are stored for additional investigations. A calibration of the system can be done with marks on the sample. The pictures are suitable to determine geometrical crack tip parameters like δ_5 or CTOA and to measure deformation fields using image correlation if a stochastic or regular pattern is applied to the sample. In all cases a fix set-up and specimen are necessary during the scan. The digital image size of 50–100 megabyte can nowadays easily be handled by actual computer equipment.

The proposed simulation method in the improved program *PCCS-2D* provides a robust concept to carry out flaw assessment with the FAD procedure in combination with a numerical crack path simulation. The experimental crack paths and the simulated ones are in good accordance. Additional initiating cracks can be considered by an easy geometry update. In the case of limit load failure the program gives a conservative prediction of the critical crack length resulting from the conservative model for calculation of limiting curve from [23].

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